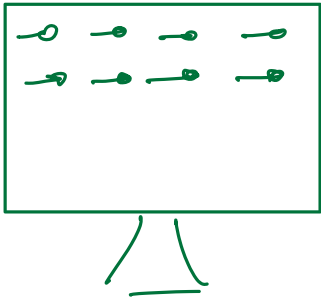


2D Arrays



	0	
1	1	0

Ludo, chess, score board, tic-tac-toe

Syntax

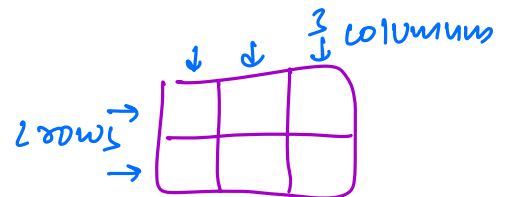
int mat [5][4] = new int [5][4]

no. of rows no. of columns

1D array :

```
int arr[] = new int[5]
```

```
int [][] mat = new int [row] [col]
```



A 3x5 grid representing a 2D array. The columns are labeled 0, 1, 2, 3, 4 and the rows are labeled 0, 1, 2. The cell at row 1, column 2 is highlighted in yellow. A red line points from this cell to the cell at row 2, column 0, which is also highlighted in yellow.

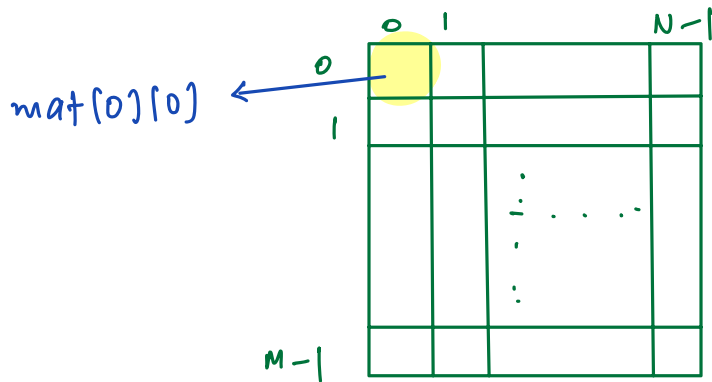
row = 1, col = 2
mat[1][2]

mat[2][0] 3x5 matrix

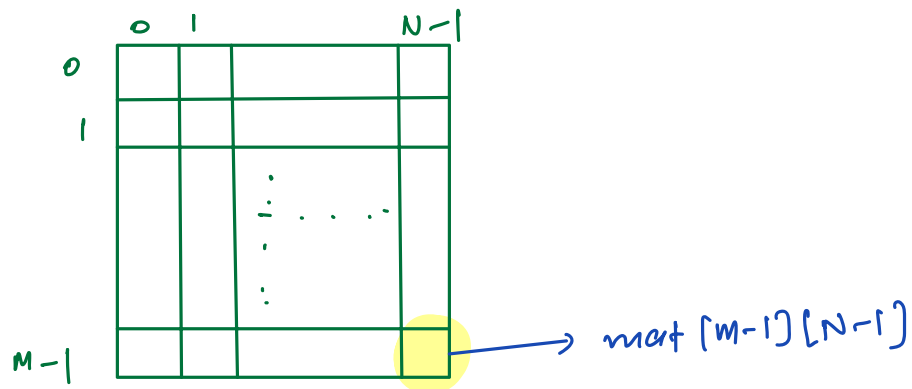
Quiz 1: 5 cols & 7 rows

`int mat[7][5] = new int[7][5]`

Quiz 2: $M \times N$ matrix, top-left corner?



Quiz 3: $M \times N$ matrix, bottom-right corner?



Question 1

Given a matrix of size $N \times M$, print its top row.

	0	1		$M-1$
0				
1				
			$\vdots$	
			$\vdots$	
			$\vdots$	
$N-1$				

top-row: $(0,0), (0,1), (0,2), \dots, (0, M-1)$

```
for( int col=0; col < M; col++) {  
    print( mat[0][col]);  
}
```

3

Question 2:

Given a matrix of size $N \times M$, print its leftmost column.

leftmost column : $(0,0)$
 $(1,0)$
 $(2,0)$
 $\vdots$
 $(N-1,0)$

```
for( int row=0; row < N; row++) {  
    print (mat[row][0]);  
}
```

Question 3 :

Given a matrix of size $N \times M$, print row by row.

```
for( int col=0; col < M; col++) {  
    print (mat[0][col]);
```

→ prints 1st row

```
}
```

```
for( int col=0; col < M; col++) {  
    print (mat[1][col]);
```

→ prints 2nd row

```
}
```

```

for (int row = 0; row < N; row++) {
    for (int col = 0; col < M; col++) {
        print(mat[row][col]);
    }
    println(); → prints newline
}

```

Question 4:

Given a matrix of size $N \times M$, print column by column.

1	3	-2
9	0	8

o/p →

```

1  9
3  0
-2 8

```

```

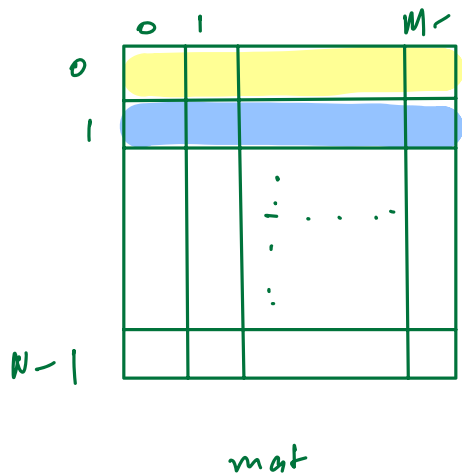
for (int row = 0; row < N; row++) {
    print(mat[row][0]); → prints 1st column
}

```

```

for (int col = 0; col < M; ++col) {
    for (int row = 0; row < N; ++row) {
        print (mat[row][col]);
    }
}

```



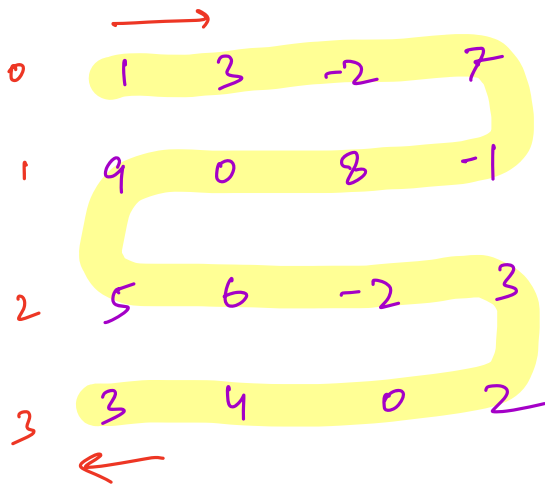
$\text{mat.length} \Rightarrow$ no. of rows

$\text{mat}[0].length \Rightarrow$ no. of columns

$\text{mat}[1].length \Rightarrow$ no. of columns

Question

Given a matrix of size $N \times M$, print it in waveform.



o/b →

1	3	-2	7
-1	8	0	9
5	6	-2	3
2	0	4	3

Solution :

print even-indexed rows in regular way
print odd-indexed rows in reverse order

```
for (int row = 0; row < N; row++) {  
    if (row % 2 == 0) { → even indexed row  
        for (int col = 0; col < M; col++) {  
            print(mat[row][col]);  
        }  
    }  
    else { → odd indexed row
```

```

        for (int col = m-1; col >= 0; col--) {
            print(mat[row][col]);
        }
    }
    println();
}

```

Question

Given a matrix of size $N \times M$, return a row-wise sum.

1 3 -2 7	$\xrightarrow{\text{sum}}$ 9	o/p : [9 16 12]
9 0 8 -1	$\rightarrow$ 16	
5 6 -2 3	$\rightarrow$ 12	

```

int[] rowWiseSum (int mat[][]) {

```

```

    int n = mat.length;

```

```

    int m = mat[0].length;

```

```

    int ans[] = new int[n];

```

```

for (int row = 0; row < n; ++row) {
    int sum = 0;
    for (int col = 0; col < m; ++col) {
        sum += mat[row][col];
    }
    ans[row] = sum;
}

return ans;
}

```

Question

Given a matrix of size $N \times M$, return column wise max.

1	3	-2	7
9	0	8	-1
5	6	-2	3
↓	↓	↓	↓
9	6	8	7

o/p $\rightarrow [9 \ 6 \ 8 \ 7]$

```
int[] colWiseMax( int mat[][]) {
```

```
    int n = mat.length;
```

```
    int m = mat[0].length;
```

```
    int ans[] = new int[m];
```

```
    for (int col = 0; col < m; ++col) {
```

```
        int max = mat[0][col];
```

```
        for (int row = 0; row < n; ++row) {
```

```
            if (max < mat[row][col])
```

```
                max = mat[row][col];
```

```
        }
```

```
        ans[col] = max;
```

```
    }
```

```
    return ans;
```

```
}
```

https://www.scaler.com/topics/java/online-java-compiler/?snippet_slug=fec1d67e7547d254fa68